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PROCEDURE

- The Vernier caliper is checked for zero errors and error if found is to be noted
- The sphere is kept in between the jaws of the Vernier caliper and the main scale reading (MSR)is noted.
- Vernier scale division which coincides with some main scale division (VSD) is noted. Zero correction made with this VSD gives Vernier scale reading (VSR).
- Multiply this VSR by Least Count (LC) and add it with MSR. This will be the diameter of the sphere.
- Observations are to be recorded for different positions of the sphere and the average value of the diameter is found. From this value radius of the sphere R is calculated.
- Using the known value of the mass of the sphere M and calculated radius of the sphere R the moment of inertia of the given sphere about its diameter can be calculated using the given formula.

LEAST COUNT (LC)

One main scale division (MSD) = ... cm

Number of Vernier scale divisions = ...

$$\text{Least Count (LC)} = \frac{\text{1 Main Scale Division (MSD)}}{\text{Total Vernier scale divisions}}$$

= ..... cm

OBSERVATIONS

Zero error =

Zero correction =

Sl.No.	MSR cm	Vernier coincidence VSD	VSR = (VSD ± ZC)	Diameter of the sphere = 2R = (MSR +VSR x LC) cm
1				
2				
3				
4				
5				
6				

Mean diameter 2R = ... cm

Radius of the sphere R = ... cm

R = ... m